

§5.1

Probability Distributions and Random Variables

Assigning numerical values to the outcomes of probability experiments

Today's Goal

After completing this section, you should be able to:

- Define random variables and distinguish between discrete and continuous random variables
- Understand the requirements for a discrete probability distribution
- Represent a discrete probability distribution using a table or histogram
- Calculate probabilities from a discrete probability distribution

Learning Target 8

LT.8 Discrete Probability Distributions

I understand what a discrete probability distribution is (including binomial distributions). I can represent the discrete probability distribution of a given random variable by creating an appropriate table or graph and can use it to calculate the probability of an event. I can also calculate the mean, variance, and standard deviation for a discrete probability distribution.

One Student, Chosen at Random

We record the number of classes the student is taking this semester, and the student's favorite color.

Which one is a numerical value, determined by chance?

Random Variables

Definition 5.1.1

A *random variable* is a variable that has a single numerical value, determined by chance, for each outcome of a probability experiment. A random variable is typically represented by a capital letter (such as X), while a lowercase letter (such as x) represents a specific value of that random variable.

Two Types of Random Variables

Random variables can be classified into two types.

Definition 5.1.2

A *discrete random variable* is a random variable that has a finite or countable number of possible values.

Definition 5.1.3

A *continuous random variable* is a random variable that has infinitely many values that are not countable. These values often form an interval on the number line.

Identify the Type of Variable

One student is chosen at random from a statistics course. Identify each of the following as a discrete random variable, continuous random variable, or not a random variable.

- (a) The number of text messages the chosen student sent yesterday.

Numbers of text messages are whole numbers, so they are countable.

This is a discrete random variable.

- (b) The exact weight of the chosen student's backpack.

Weight can take any value within a range (e.g., 12.6481 . . . pounds). The values are not countable. This is a continuous random variable.

Identify the Type of Variable

One student is chosen at random from a statistics course. Identify each of the following as a discrete random variable, continuous random variable, or not a random variable.

(c) The class level of the chosen student.

The values for this variable (freshman, sophomore, etc.) are not numerical. This is not a random variable.

Now You Try: Types of Variables

You Try 5.1.1. A statistics class takes an exam. One student is chosen at random from the class. Identify each of the following as a discrete random variable, continuous random variable, or not a random variable.

- (a) The number of exam questions the chosen student answered correctly.
- (b) The time it took the chosen student to complete the exam.
- (c) The eye color of the chosen student.

You Try 5.1.1: Checking Your Answer

(a) The number of exam questions the chosen student answered correctly.

Discrete random variable. The values are whole numbers, so they are countable.

(b) The time it took the chosen student to complete the exam.

Continuous random variable. Time can take any value within a range.

(c) The eye color of the chosen student.

Not a random variable. The values are not numerical.

For this chapter, we will focus on discrete random variables. Continuous random variables will be studied in Chapter 6.

Discrete Probability Distributions

Definition 5.1.5

A *discrete probability distribution* (also called a discrete probability distribution function or discrete PDF) is a function that assigns a probability to each value of a discrete random variable.

Discrete probability distributions can be represented using tables, formulas, or graphs.

Three Requirements

A discrete probability distribution P must satisfy the following requirements:

1. There is a discrete random variable X , and each value x is associated with a probability $P(x)$.
2. Each probability is between 0 and 1, inclusive: $0 \leq P(x) \leq 1$ for every value of x .
3. The sum of all probabilities equals 1: $\sum P(x) = 1$.

Is This a Probability Distribution?

College students were asked which study methods they use before an exam, and the table below summarizes their responses. Does this table describe a probability distribution?

x	$P(x)$
Rereading notes	0.61
Flash cards	0.48
Practice problems	0.37
Study groups	0.22

Is This a Probability Distribution?

x	$P(x)$
Rereading notes	0.61
Flash cards	0.48
Practice problems	0.37
Study groups	0.22

This is **not** a discrete probability distribution for two reasons:

1. The values of the variable are not numerical (they are categories like “Rereading notes”). So X is not a random variable.
2. The probabilities add up to $0.61 + 0.48 + 0.37 + 0.22 = 1.68$, which is greater than 1.

Number of Heads in Three Flips

A fair coin is flipped three times. Let X represent the number of heads obtained. Find the probability distribution for X .

Since we are flipping the coin three times, the number of heads can be 0, 1, 2, or 3.

The sample space for this experiment is

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

Counting the Outcomes

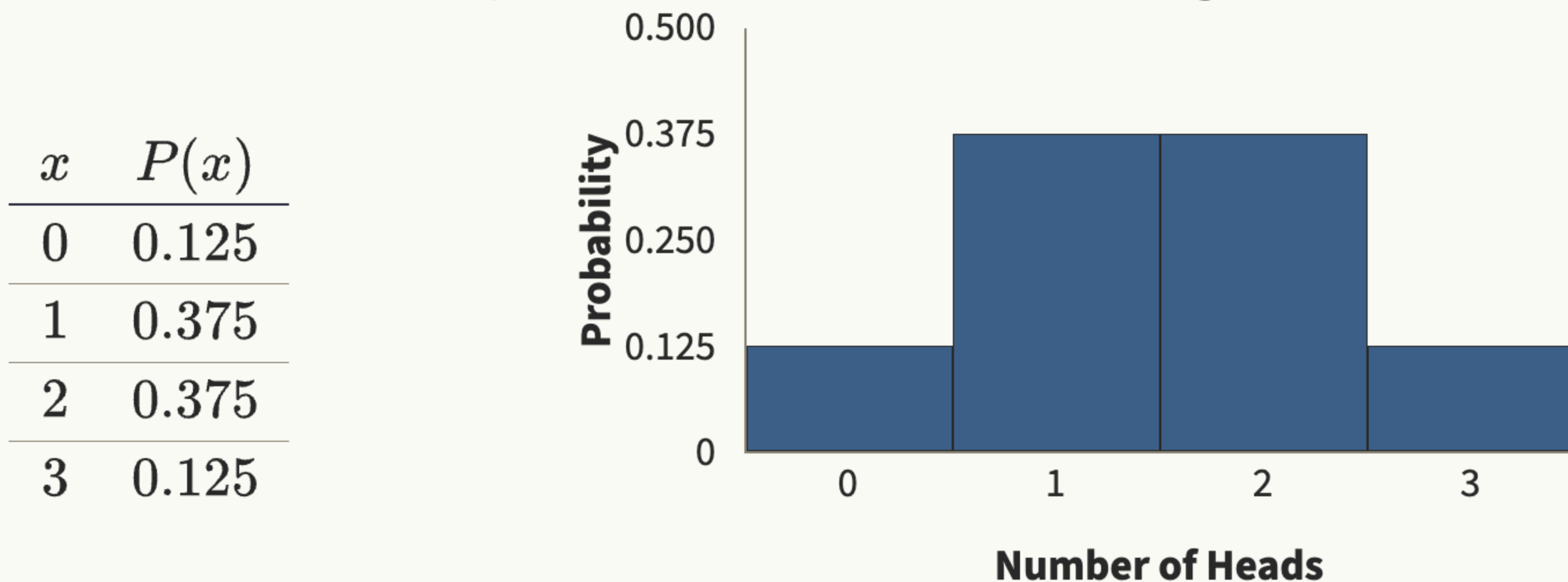
$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

Counting the outcomes for each value of X :

- $P(X = 0)$: Only TTT has 0 heads, so $P(0) = \frac{1}{8} = 0.125$.
- $P(X = 1)$: HTT, THT , and TTH have 1 head, so $P(1) = \frac{3}{8} = 0.375$.
- $P(X = 2)$: HHT, HTH , and THH have 2 heads, so $P(2) = \frac{3}{8} = 0.375$.
- $P(X = 3)$: Only HHH has 3 heads, so $P(3) = \frac{1}{8} = 0.125$.

A Table and a Histogram

We can represent this probability distribution as a table or a histogram:



Notice that the probabilities sum to $0.125 + 0.375 + 0.375 + 0.125 = 1$, as required.

The Sum of Two 4-Sided Dice

A pair of fair 4-sided dice is rolled. Let X represent the sum of the two numbers rolled.

(a) Construct a table representing the probability distribution for X .

The sample space has 16 equally likely outcomes. Each cell below shows the sum:

+	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

Counting Each Sum

+	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

Counting how many times each sum appears, we get the probability distribution:

x	2	3	4	5	6	7	8
$P(x)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

A Sum of at Least 6

x	2	3	4	5	6	7	8
$P(x)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

(b) Find $P(X \geq 6)$. Round to 4 decimal places.

Add the probabilities for $x = 6, 7, 8$:

$$P(X \geq 6) = P(6) + P(7) + P(8)$$

$$= \frac{3}{16} + \frac{2}{16} + \frac{1}{16} = \frac{6}{16} = 0.3750.$$

An Odd Sum

x	2	3	4	5	6	7	8
$P(x)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

(c) Find the probability that X takes an odd value. Round to 4 decimal places.

Add the probabilities for $x = 3, 5, 7$:

$$P(X \text{ is odd}) = \frac{2}{16} + \frac{4}{16} + \frac{2}{16}$$

$$= \frac{8}{16} = 0.5000.$$

Now You Try: A Missing Probability

You Try 5.1.2. A campus boathouse checks each kayak as it comes off the water and marks any fresh scrapes along its hull. One of these kayaks is chosen at random. Let X be the number of fresh hull scrapes on the chosen kayak.

x	0	1	2	3	4
$P(x)$	0.45	0.25	?	0.10	0.05

Now You Try: A Missing Probability

You Try 5.1.2, continued.

x	0	1	2	3	4
$P(x)$	0.45	0.25	?	0.10	0.05

Round each answer to two decimal places.

- (a) Find the missing probability, $P(X = 2)$.
- (b) Find $P(X \geq 2)$.
- (c) Find the probability that the chosen kayak has at least one fresh hull scrape.

You Try 5.1.2: Checking Your Answer

x	0	1	2	3	4
$P(x)$	0.45	0.25	?	0.10	0.05

(a) The sum of all probabilities equals 1.

The four probabilities in the table add to 0.85, so

$$P(X = 2) = 1 - 0.85 = 0.15.$$

(b) Add the probabilities for $x = 2, 3, 4$.

$$P(X \geq 2) = 0.15 + 0.10 + 0.05 = 0.30.$$

(c) At least one means every value of X except 0.

$$1 - P(X = 0) = 1 - 0.45 = 0.55.$$

Probability Histograms

A *probability histogram* is a histogram where the horizontal axis represents the values of the random variable and the vertical axis represents the probabilities.

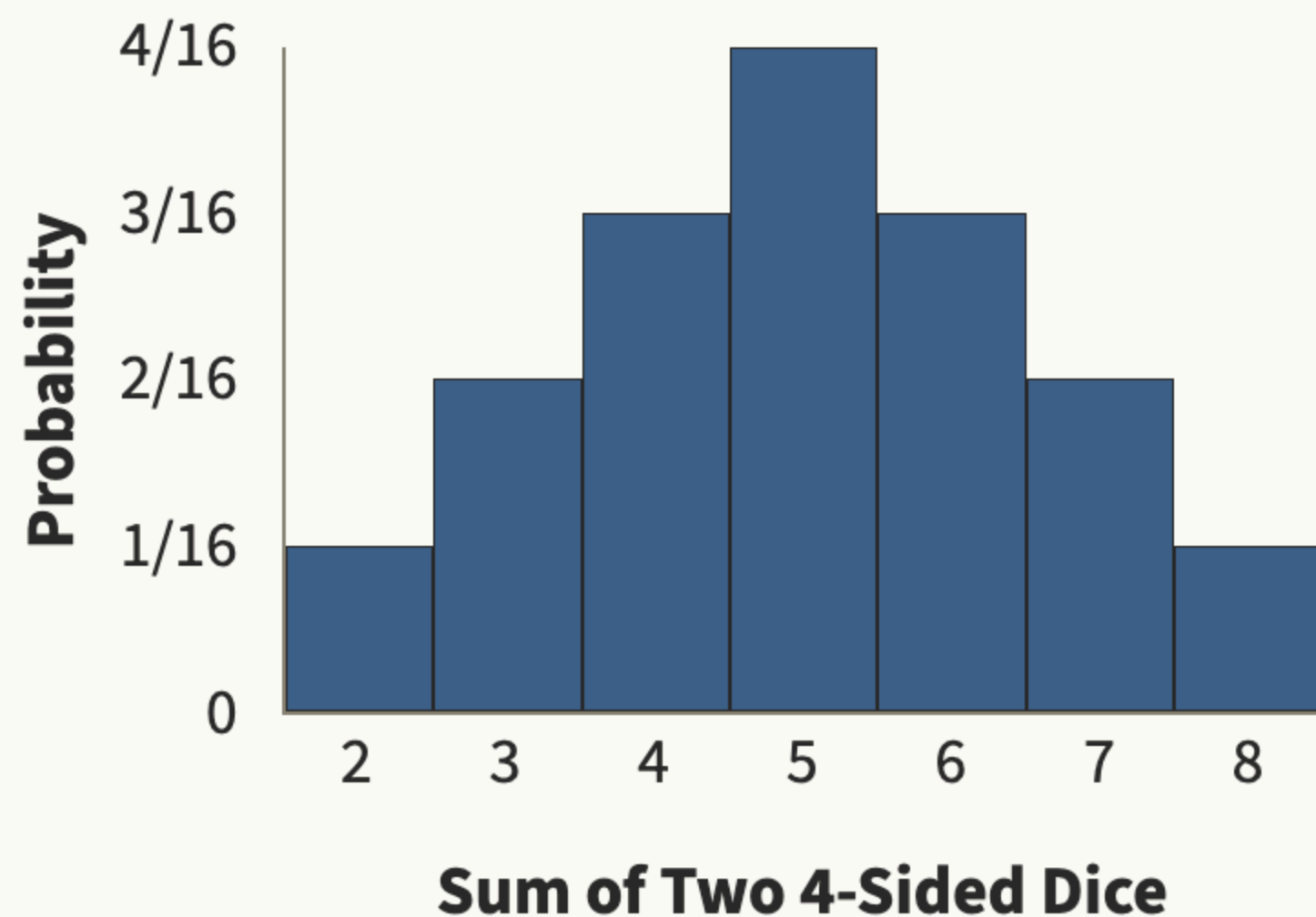
Unlike frequency histograms, probability histograms have bars whose heights (or areas) represent probabilities rather than frequencies.

Here is the probability distribution for the sum X of two fair 4-sided dice:

x	2	3	4	5	6	7	8
$P(x)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

Sketch the probability histogram for this distribution.

The Shape of a Probability Histogram



The shape of a probability histogram tells us about the distribution:

- This histogram is *symmetric*: it looks the same on both sides of the center.
- The most likely outcome (highest bar) is $X = 5$.
- Extreme values (2 or 8) are least likely.

What to Take With You

- A random variable has a single numerical value, determined by chance, for each outcome of a probability experiment.
- A discrete random variable has a finite or countable number of possible values.
- In a discrete probability distribution, each probability is between 0 and 1, inclusive, and the sum of all probabilities equals 1.
- A discrete probability distribution can be represented using a table or a histogram.
- To find the probability of several values of X , add their probabilities.